ZHAOHUI LI

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EMPLOYMENT

- **College of AI, Cyber and Computing, The University of Texas at San Antonio** *Sep 2026 – Present*
  Postdoctoral Associate | *Mentor: Dr. Jinjun Xiong* San Antonio, TX
- **Institute for Artificial Intelligence and Data Science (IAD), Department of Computer Science and Engineering (CSE), University at Buffalo** *Aug 2024 – Aug 2026*
  Postdoctoral Associate | *Mentor: Dr. Jinjun Xiong* Buffalo, NY
  - AI Team Lead at the **National AI Institute for Exceptional Education (AI4ExceptionalEd)**.
  - AI Team Lead at the **Center for Early Literacy and Responsible AI (CELaRAI)**.

EDUCATION

- **The Pennsylvania State University** *Aug 2019 – May 2024*
  Ph.D. in Computer Science — Natural Language Processing Lab State College, PA
  - **Advisor:** Dr. *Rebecca J. Passonneau*
  - **Dissertation:** *Automated and Semi-Automated Assessment of STEM Reasoning Questions* (***Best Doctoral Dissertation Award***)
- **Nankai University** *Aug 2012 – Jun 2016*
  B.S. in Computer Science and Engineering Tianjin, China
  - **Advisors:** *Dr. Jun Xu* & *Dr. Jie Liu*
  - **Thesis:** BDA: Big Data Analysis as a Service (2k+ GitHub Stars)

GRANTS EXPERIENCE

- **Organization for Autism Research (OAR) Research Grant ($50,000)** *Feb 2026 – Dec 2027*
  ***Co-PI**; Study of Parent's AI Coach for Parent-Implemented Communication Interventions for Children with ASD.*
- **NSF #2229873 — AI Institute for Exceptional Education ($20,000,000)** *Jan 2023 – Dec 2027*
  Tech Lead supporting Dr. Jinjun Xiong (Co-PI); advancing AI solutions for challenges in special education.
- **Penn State AI Challenge Grant — Nittany AI Challenge 2023 ($5,000)** *Jan 2023 – Jan 2024*
  ***PI**; Developed an AI-driven Interactive E-Books System for Children.*

SELECTED RESEARCH EXPERIENCE

- **SciEval: Evaluating K–12 Science Instructional Materials** *July 2025–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong*
  Collaborators: Profs. Tingting Li and Peng He, Washington State University
  - Lead task formulation, rubric co-design, and model fine-tuning for evaluating science instructional materials; work with education experts to validate LLM scores, reasoning, and suggestions.
  - We are developing the first platform that help teachers to understand, evaluate, and adapt STEM instructional material in classrooms. A Proposal to NSF STEM K-12 is in preparation.
  - **Publications:** SciEval (*AIED 2026*) [3]; Judging the Judges (*Computing Conference 2026*) [5].

- **AI Reading Enhancer (AIRE): Decodable Story Generation and Evaluation** *July 2024–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong (Center Co-PI)*
  Collaborators: Prof. Tanya Christ, John Strong, and Laura Tortorelli (co-PIs)
  - Develop teacher-guided story-generation tools and lead AIRE development at the CELaRAI; co-design rubrics for phonics alignment, narrative coherence, and illustration quality with literacy experts.
  - **Funding (CELaRAI):** IES #R305C240046; $10,000,000; project period: Sep 2024–Aug 2029.
  - **Publications/presentations:** StoryLab (*AIED 2025*) [7]; AIRE (*AERA 2026 poster*) [6].

- **Embodied AI: Reading Companions for Children's AI Literacy** *Nov 2024–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong*
  Collaborator: Prof. Shiyang Jiang, University of Pennsylvania

- Implement pedagogically structured dialogue for an LLM-powered reading robot; evaluate dialogue policies using simulated child profiles and expert review.
- Collaborated on studies of bilingual conversational agents and dialogic reading, examining children's participation, learning, and engagement in AI-guided and parent-guided reading.
- **Publication:** Reading companion for children's AI literacy (*AIED 2026 Companion Proceedings*) [2]; bilingual conversational reading (*Computers and Education: Artificial Intelligence*, 2025) [9]; parent-led versus AI-guided reading (*BJET*, 2025) [10].

- **PaiCoach: AI Assessment and Coaching for Autism Home Interventions** *Nov 2024–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong*
  - Lead ASD-HI benchmark and PaiCoach AI model development as part of the AI4EE project, assessing parent–child interactions and generating feedback for specialist review.
  - We recently submitted the latest dataset for benchmarking autistic children speech recognition to LAK 2027, and a related grant proposal to NSF Future CoRe is in preparation.
  - Serve as Co-PI on the $50,000 OAR-funded PAC.AI project, extending this work to AI-supported parent coaching.
  - **Publications:** ASD-HI (*AIED 2025*) [8]; PaiCoach (*Journal of Autism and Developmental Disorders*, 2026) [1].

- **LLM4Art: Expert-Guided Art Interpretation and Evaluation** *July 2025–Present*
  Research Lead | *Mentor: Prof. Jinjun Xiong*
  Collaborator: Prof. Kyunghee Pyun, Fashion Institute of Technology
  - Lead application and evaluation development, translating expert knowledge into prompts and rubrics for visual analysis, historical context, and cultural significance.
  - **Publication:** LLM4Art (*LAK 2026*) [4].

- **Automated Formative Assessment of STEM Short Response** *Aug 2019–May 2024*
  Research Lead | *Mentor: Prof. Rebecca J. Passonneau, The Pennsylvania State University*
  Collaborators: Prof. Matthew D. Beckman, The Pennsylvania State University; Prof. Sadhana Puntambekar, University of Wisconsin–Madison
  - Developed SFRN and AsRRN to model relationships among questions, reference responses, and student answers; investigated demonstration examples for LLM assessment of science essays.
  - **Publications:** SFRN (*EMNLP 2021*) [[19]]; AsRRN (*Findings of EMNLP 2023*) [[13]]; science-essay assessment (*AIED 2025 workshop*) [[11]].

- **STEM Assessment Datasets and Human Scoring Rubrics** *Aug 2019–May 2024*
  Research Lead | *Mentor: Prof. Rebecca J. Passonneau, The Pennsylvania State University*
  Collaborator: Prof. Matthew D. Beckman, The Pennsylvania State University
  - Contributed to ISTUDIO and physics lab-report datasets, including rubric development and human-scoring procedures for automated assessment research.
  - **Datasets:** ISTUDIO (2023) [15]; rubric-based assessment of physics lab reports (2022) [18].

- **Human–AI Collaboration: Selective Prediction and Human Deferral** *Aug 2019–May 2024*
  Research Lead | *Mentor: Prof. Rebecca J. Passonneau, The Pennsylvania State University*
  Collaborator: Prof. Sadhana Puntambekar, University of Wisconsin–Madison
  - Developed methods for automated assessment models to answer or defer to a person, including joint optimization of assessment and deferral representations.
  - **Publications:** Human deferral for short-answer grading (*AIED 2023*) [14]; joint training for selective prediction (*arXiv preprint*, 2024) [12].

Research Interests

My research focuses on human-centered AI technologies for education that translate educational goals and expert knowledge into computational tasks. My work bridges AI and education by translating educational goals into task-specific computational goals. I have developed algorithms, benchmark datasets, and systems that support teaching and learning, including automated assessment and feedback in special education, instructional material evaluation, storybook generation, and student behavior analysis. My career goal is to develop reliable and responsible AI frameworks to promote equitable, high-quality education worldwide.

My future research will continue pursuing interdisciplinary collaborations to develop AI applications that address specific educational needs. Building on these collaborations, I will develop a theoretical framework

for translating pedagogical goals into computational tasks and an agent-based system to automate parts of task formulation and evaluation design. I will also investigate how to use AI responsibly in education, focusing on potential risks such as emotional overreliance on AI and declines in students' independent reasoning. These directions will contribute to the design of education colloquim that supports learning and human connection in the era of AI.

Keywords: Large Language Models, Natural Language Processing, Educational Assessment, Educational Technology, Early Literacy, Learning Science, AI Literacy, Art Education, Special Education

Teaching Experience

- **NSF Argumentation Workshop (PACE Lab)** *Feb 2022*
  Invited Speaker Illinois State University
  ◦ Delivered a workshop talk on AI-assisted assessment and Large Language Models.
- **Pennsylvania State University** *Jan 2020 – May 2020*
  Teaching Assistant — Natural Language Processing (CSE597, graduate level) State College, PA
  ◦ Teaching and grading; supported student projects.
- **Pennsylvania State University** *Aug 2019 – Dec 2019*
  Teaching Assistant — Artificial Intelligence (CMPSC442, undergraduate level) State College, PA
  ◦ Teaching and grading; exams for 100+ students.

Service Experience

- **Proposal Reviewer:** NSF Panelist on DRL
- **Conference Reviewer:** ACL Rolling Review(ACL, EMNLP, NAACL), AAAI, AIED, LAK, SIGIR
- **Session Chair:** AIED 2025, 2026 (Language Learning & Story Generation, AI Agents for Education)
- **Program Committee Member:** AIED 2025, AIED 2026, LAK 2027
- **Student Volunteer:** AAAI 2019

Edited Books

- Akemoğlu, Y., Xiong, J., **Li, Z.**, & Zheng, Q. (Eds.). *Human-Centered AI for Special Education – Design, Assessment, and Intervention* (provisional title). Springer. **Under contract; in preparation.**

Publications

[1] Akemoğlu, Y., **Li, Z.**, Singh, S. P., Zheng, Q., Roberts, E. B., & Xiong, J. Parent's AI Coach (PaiCoach): A Single-Case Experimental Study of an AI-Supported Parent Coaching System for Autistic Children. *Journal of Autism and Developmental Disorders*.

[2] Xiao, F., **Li, Z.**, Jiang, S., & Xiong, J. (2026). Design and Feasibility of an LLM-Powered Humanoid Robot as a Reading Companion to Support Children's AI Literacy. In *Proceedings of the International Conference on Artificial Intelligence in Education*. (AIED 2026)

[3] **Li, Z.**, He, P., Chen, Z., Liu, H., Wang, Z., Li, T., & Xiong, J. (2026). SciEval: A Benchmark for Automatic Evaluation of K–12 Science Instructional Materials. In *Proceedings of the International Conference on Artificial Intelligence in Education*. (AIED 2026)

[4] **Li, Z.**, Li, N., & Xiong, J. (2026). LLM4Art: Using Large Language Models for Art Interpretation. In *Proceedings of the International Conference on Learning Analytics & Knowledge*. (LAK 2026)

[5] He, P., **Li, Z.**, Wang, Z., Xiong, J., & Li, T. (2026). Judging the Judges: Human Validation of Multi-LLM Evaluation for High-Quality K–12 Science Instructional Materials. In *Proceedings of the Computing Conference*. (Computing Conference 2026)

[6] **Li, Z.**, Zheng, Q., Liu, D., Christ, T. M., Strong, J. Z., Tortorelli, L. S., Wang, X. C., Alwan, A., Hakkani Tur, D., & Xiong, J. (2026, April). AIRE: AI Reading Enhancer for Personalized Decodable Texts in Early Literacy. Poster presented at the *American Educational Research Association Annual Meeting*, Los Angeles, CA. (AERA 2026)

[7] **Li, Z.**; Xiao, F.; Lin, J.; Zou, X.; Zheng, Q.; Xiong, J. (2025, July). StoryLab: Empowering Personalized Learning for Children through Teacher-Guided Multimodal Story Generation. In *International Conference on Artificial Intelligence in Education* (pp. 285–292). Cham: Springer Nature Switzerland. (AIED 2025)

[8] **Li, Z.**, Akemoglu, Y., Lyu, J., Zheng, Q., & Xiong, J. (2025, July). ASD-HI: A Parent-Child Interaction Dataset for Automated Assessment of Home Intervention. In *International Conference on Artificial Intelligence in Education* (pp. 48–62). Cham: Springer Nature Switzerland. (AIED 2025)

[9] Xiao, F., **Li, Z.**, Lin, J., Zou, X., Yang, D., Zou, E. W., & Xiong, J. (2025). Leveraging an LLM-enhanced bilingual conversational agent for EFL children's dialogic reading: Insights from children, parents, and educators. *Computers and Education: Artificial Intelligence*, 100484.

[10] Xiao, F., Zou, E. W., Lin, J., **Li, Z.**, & Yang, D. (2025). Parent-led vs. AI-guided dialogic reading: Evidence from a randomized controlled trial in children's e-book context. *British Journal of Educational Technology*, 56(5), 1784–1813.

[11] Sheikhi_Karizaki, M., **Li, Z.**, & Passonneau, R. J. (2025, July). Criteria for Demonstration Examples for LLM Assessment of Science Essays. In *Workshop on Epistemics and Decision-Making in AI-Supported Education, Annual Conference on Artificial Intelligence in Education*. (AIED 2025)

[12] **Li, Z.**, & Passonneau, R. J. (2024). Joint Training for Selective Prediction. *arXiv preprint* arXiv:2410.24029.

[13] **Li, Z.**, Lloyd, S. E., Beckman, M. D., & Passonneau, R. J. (2023, December). Answer-state recurrent relational network (AsRRN) for constructed response assessment and feedback grouping. In *The 2023 Conference on Empirical Methods in Natural Language Processing*. (EMNLP 2023)

[14] **Li, Z.**, Zhang, C., Jin, Y., Cang, X., Puntambekar, S., & Passonneau, R. J. (2023, June). Learning when to defer to humans for short answer grading. In *International Conference on Artificial Intelligence in Education* (pp. 414–425). Cham: Springer Nature Switzerland. (AIED 2023)

[15] **Li, Z.**; Lloyd, S.; Beckman, M. D.; Passonneau, R. J. (2023). *ISTUDIO: A Statistical Automatic Response Assessment Dataset*. Penn State Data Commons.

[16] Passonneau, R., Koenig, K., **Li, Z.**, & Soddano, J. (2023). The ideal versus the real deal in assessment of physics lab report writing. *European Journal of Applied Sciences*, 11(2).

[17] Lloyd, S., Beckman, M. D., Pearl, D. K., Passonneau, R. J., **Li, Z.**, & Wang, Z. (2022, September). Foundations for AI-Assisted Formative Assessment Feedback for Short-Answer Tasks in Large-Enrollment Classes. In *Proceedings of the 11th International Conference on Teaching Statistics (ICOTS 11)*. DOI: 10.52041/iase.icots11.T3C3.

[18] **Li, Z.**; Atil, B.; Koenig, K.; Passonneau, R. J. (2022). *Reliable Rubric-Based Assessment of Physics Lab Reports: Data for Machine Learning*. Penn State Data Commons.

[19] **Li, Z.**, Tomar, Y., & Passonneau, R. J. (2021, November). A semantic feature-wise transformation relation network for automatic short answer grading. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing* (pp. 6030–6040). (EMNLP 2021)

[20] **Li, Z.**, Feng, Y., Xu, J., Guo, J., Lan, Y., & Cheng, X. (2019, July). Teaching machines to extract main content for machine reading comprehension. In *Proceedings of the AAAI Conference on Artificial Intelligence* (Vol. 33, No. 1, pp. 9973–9974).

[21] **Li, Z.**, Xu, J., Lan, Y., Guo, J., Feng, Y., & Cheng, X. (2018, September). Hierarchical answer selection framework for multi-passage machine reading comprehension. In *China Conference on Information Retrieval* (pp. 93–104). Cham: Springer International Publishing. (**Best Paper Nominee**)

[22] Guo, T., Xu, J., Yan, X., Hou, J., Li, P., **Li, Z.**, . . . & Cheng, X. (2016, October). Ease the process of machine learning with dataflow. In *Proceedings of the 25th ACM International Conference on Information and Knowledge Management (CIKM 2016)* (pp. 2437–2440). (**Best Demo Paper Nominee**)

[23] Wang, Y.; **Li, Z.**; Liu, J.; He, Z.; Huang, Y.; Li, D. (2014). *Word Vector Modeling for Sentiment Analysis of Product Reviews*. NLPCC 2014.